

Elijah Croft

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EDUCATION

California State University, Fullerton

Fullerton, CA

B.S. Computer Science & B.A. Philosophy (Double Major)

Expected May 2028

- Undergraduate research in AI ethics; presented a panel on fairness, accountability, and bias in algorithmic decision-making.

EXPERIENCE

Software Engineer (Contract)

May 2025 – Present

2E4ME

Newport Beach, CA

- Shipped the full-stack scheduling platform now coordinating **20+ instructors and 70+ students**, retiring a manual spreadsheet process that cost staff hours every term; React 19 + TypeScript on an Express REST API.
- Built the scheduling core as an **OptaPlanner** constraint solver, resolving availability, room capacity, and enrollment into conflict-free timetables in seconds, on multi-tenant infrastructure secured with **Google OAuth 2.0 SSO**.

Robotics Coach & Programming Instructor

Mar 2025 – Present

Magikid Robotics Lab

Tustin, CA

- Led a team of 10 to two **VEX AI World Championship** finishes (18th and 8th against international fields) and a first-place win at the VEX AI International Open at UC Berkeley, owning code review and autonomous strategy.
- Drove the autonomous stack: a **Markov Decision Process** policy cutting path-planning errors **80%**, fed by **YOLOv11** detection.

PROJECTS

Mini Motorways Route Optimizer | Python, NumPy, Pillow | [GitHub](#)

2026

- Modeled house-to-store assignment as a **min-cost flow** (successive shortest paths, Johnson potentials) under per-store capacity caps, building the cost matrix in S Dijkstra sweeps instead of $H \times S$ — provably optimal, not greedy.
- Routed roads with a **Steiner-forest heuristic** over free existing tarmac, then rip-up-and-reroute and reassignment passes to a fixed point: **42–84% fewer road tiles**, traffic-shaded so chokepoints surface; a screenshot-to-board vision pipeline re-plans the live game.

Agent Poker — LLM Agents Under Imperfect Information | Python, unittest — no runtime dependencies | [GitHub](#)

2026

- Reproducible harness for LLM agents under imperfect information: no-limit **Hold'em** with side pots, plus Kuhn poker for its exact equilibrium.
- Scored policies by **exact best-response search** over the game tree, not simulation; **182 tests**.

evosim — Individual-Based Evolution Simulator | Python, NumPy, pytest

2026

- Wrote a **7,700-line agent-based simulation** in which each organism is a discrete individual with a diploid genome, resolving energy, foraging, predation, and reproduction per tick across a world of depth-attenuated light, thermal gradients, and nutrients.
- Detected speciation automatically and recorded full phylogeny; runs reproduce bit-for-bit from (*config*, *seed*) under **19 test modules**.

DormDash — Autonomous Dorm Delivery Robot | C, Python, Zephyr RTOS, ROS2, Raspberry Pi 5, Zenoh,

Fall 2025

CAD | [GitHub](#)

- Autonomous hallway delivery robot: a **ROS2** graph on a Raspberry Pi 5 over **Zephyr RTOS** firmware in C, closed-loop **PID** motor control.
- Bridged MCU and Pi over **Zenoh**, designed the 12V Li-ion power path and CAD chassis, and shipped a Next.js delivery queue.

Headless Session Monitor & Alert Service | Python, Playwright, GraphQL, Raspberry Pi, systemd

Jul 2026

- Reverse-engineered a private **GraphQL** API from live traffic; a **systemd** unit on a Pi 4 polls it and pushes mobile alerts.

IQDraw | Python, GitHub Actions | [GitHub](#) · [Docs](#)

Aug 2026

- Generated step-by-step VEX IQ build booklets from a text spec: parts at hole coordinates in, isometric step diagrams out.

PPP Simulator | Best Use of ArcGIS, FullyHacks 2026 | Next.js, TypeScript, ArcGIS SDK | [Demo](#) · [GitHub](#)

Apr 2026

- Browser ocean-drift simulator advecting spills through real **ArcGIS** ocean-current vector fields, with marine-hazard overlays.

MixSafe | Next.js, TypeScript, Tailwind CSS, Django | [Live](#) · [Design](#)

Oct 2024 – Present

- Built and deployed a harm-reduction platform serving **500+ weekly users** and ~2,000 monthly interactions, on a structured interaction database backing real-time lookup of substance combinations and their effects.
- Designed the launch campaign solo — logo system, **23 campus flyers**, and 12 rounds of press-ready stickers — each sheet carrying its own QR code, so every scan is attributable to the poster that earned it.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript/JavaScript, Java, C, Scala, SQL

Frameworks: React, Next.js, Node/Express, Django, Tailwind CSS, pytest

ML & Modeling: NumPy, YOLOv11, Markov Decision Processes, agent-based simulation, combinatorial optimization

Systems & Embedded: ROS2, LiDAR SLAM & localization, Raspberry Pi 4/5, Jetson Orin Nano, Li-ion power & BMS, CAD, VEX V5/IQ

Tools: Git, Docker, Linux, Bash, Make, GitHub Actions, Playwright, REST & GraphQL APIs